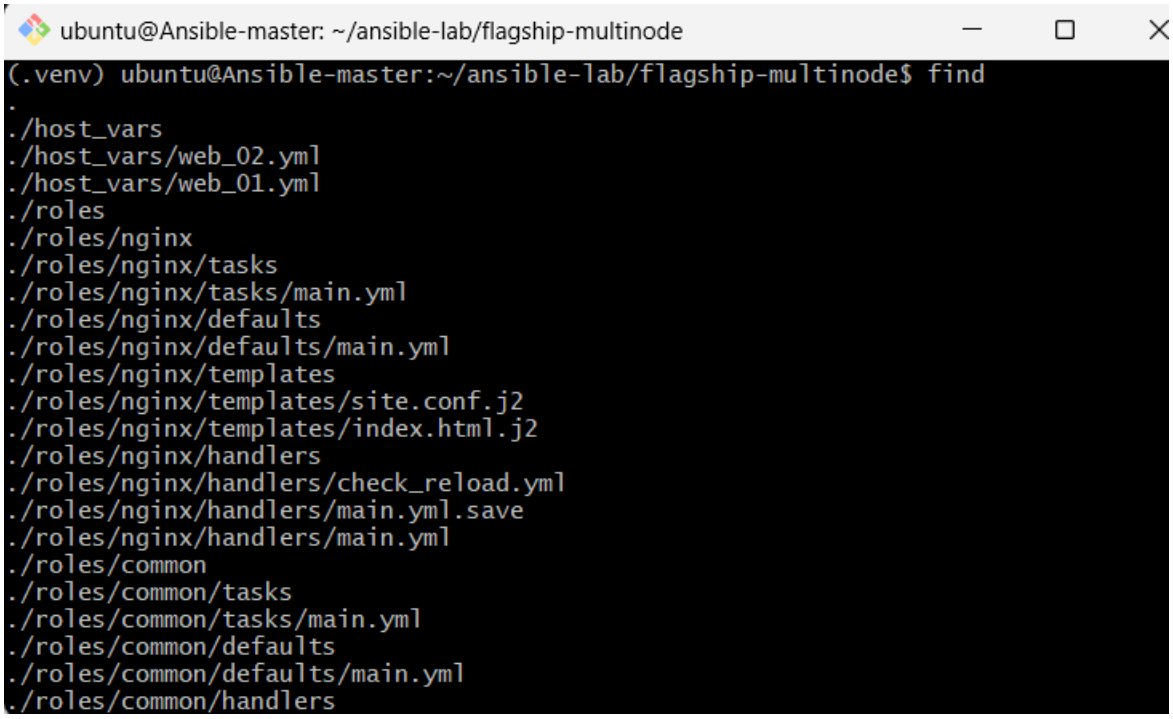
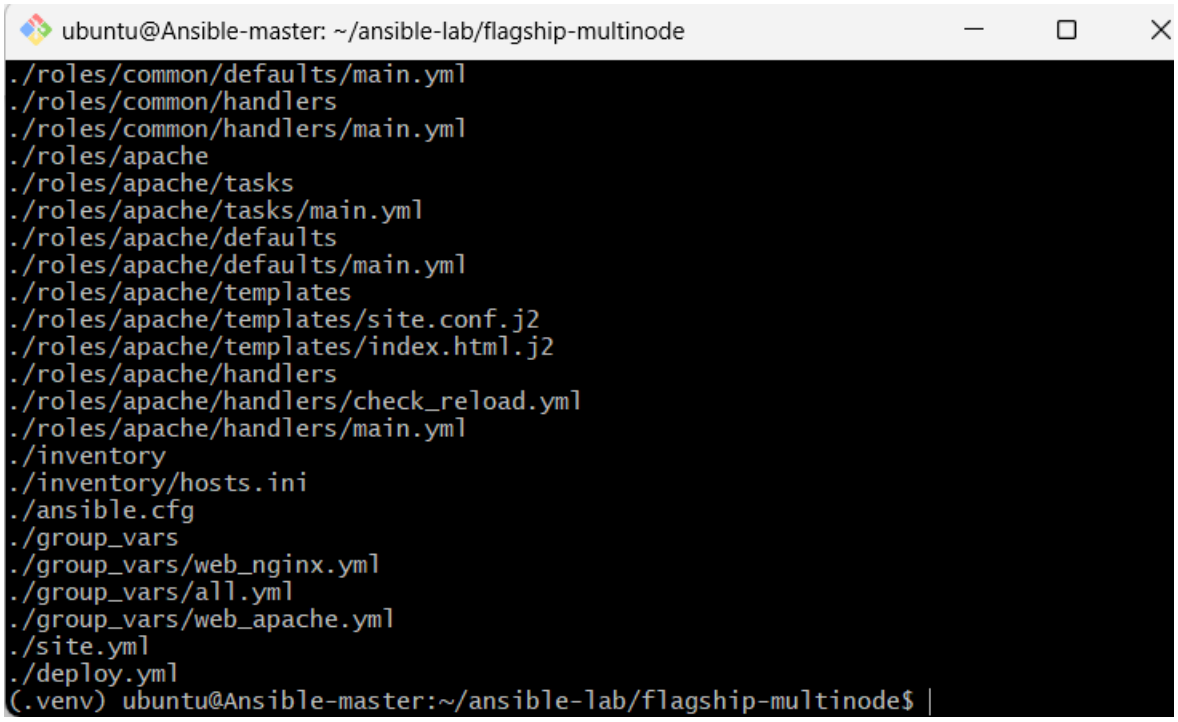
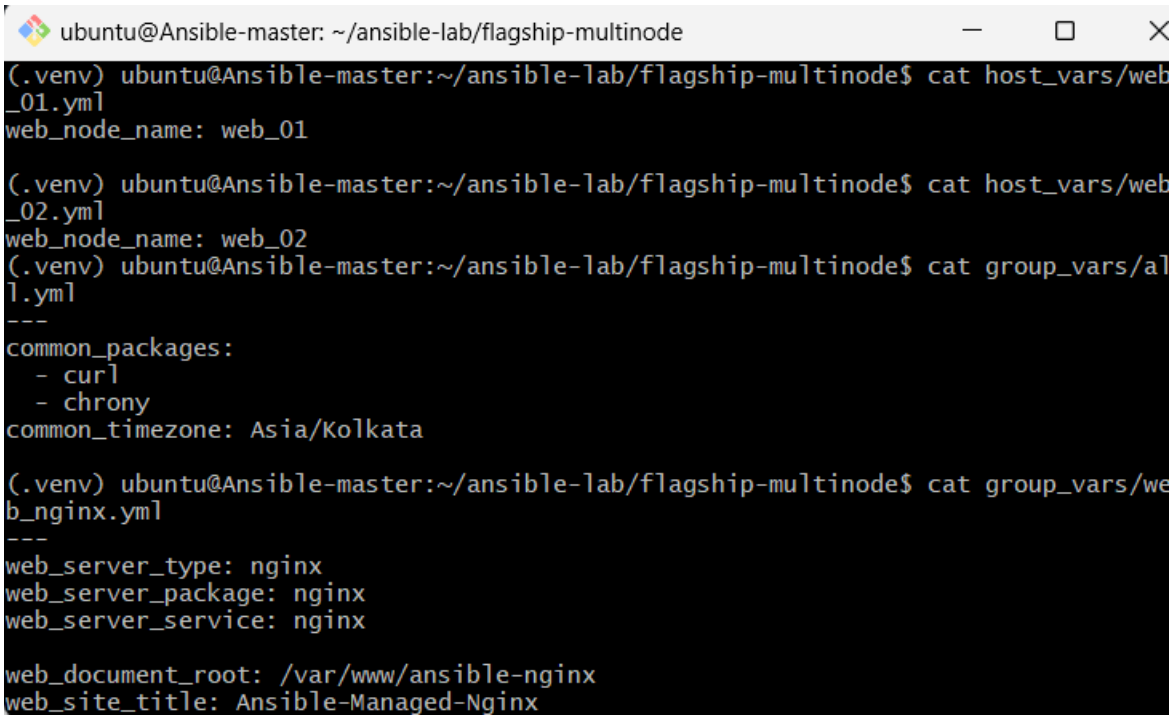
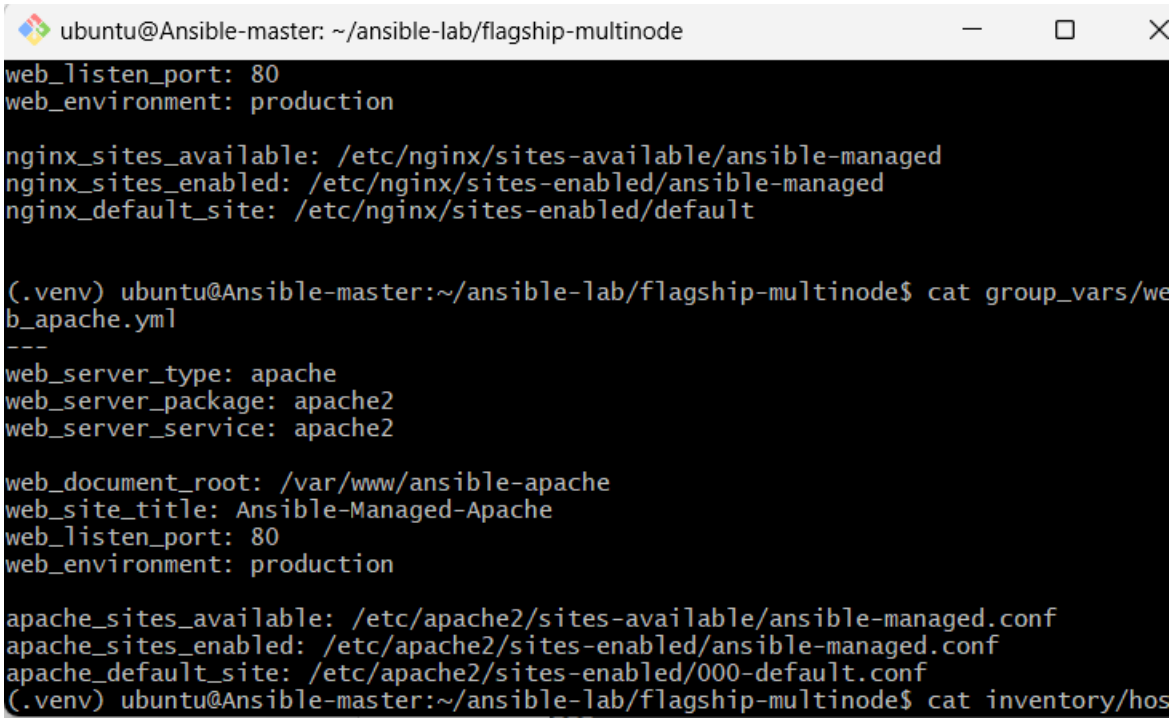
Ansible Mastery Part-6(Flagship Practical-2)

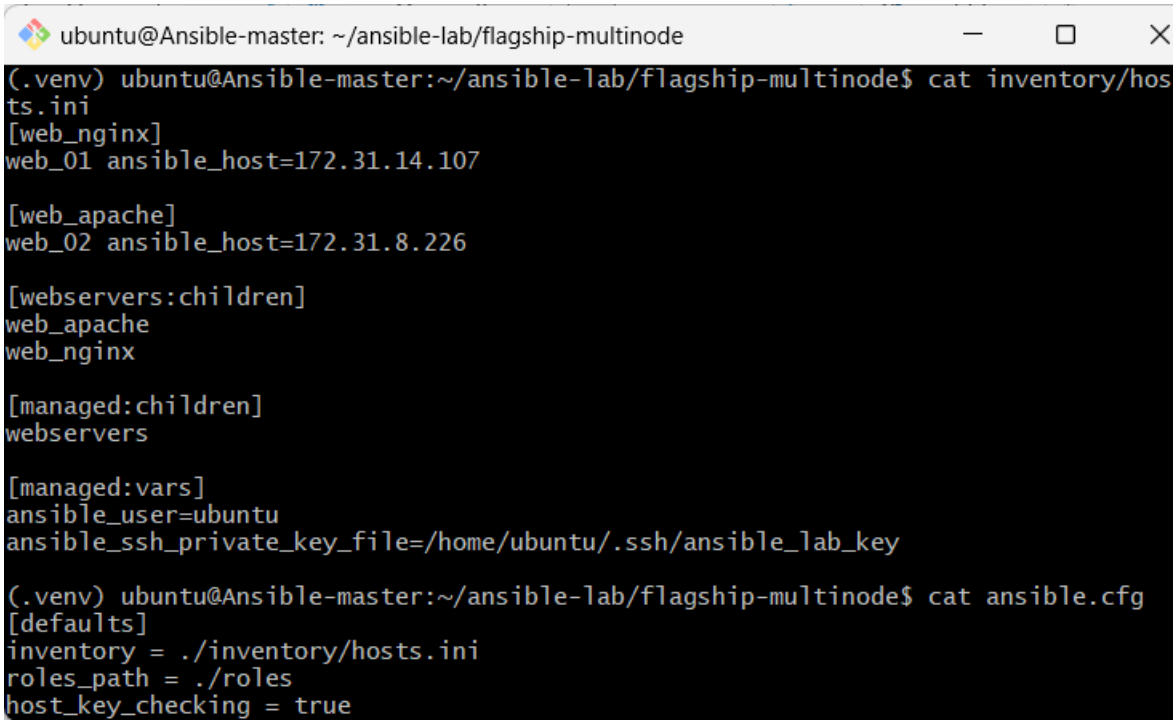
Enterprise-Grade Multi-Node Infrastructure Orchestration and Configuration Management: Engineering Reusable Role-Based Automation, Shared System Baselines, Heterogeneous Nginx and Apache Web Server Deployments, Declarative Configuration, Service Lifecycle Management, and Automated Health Verification Across Distributed Ubuntu Hosts

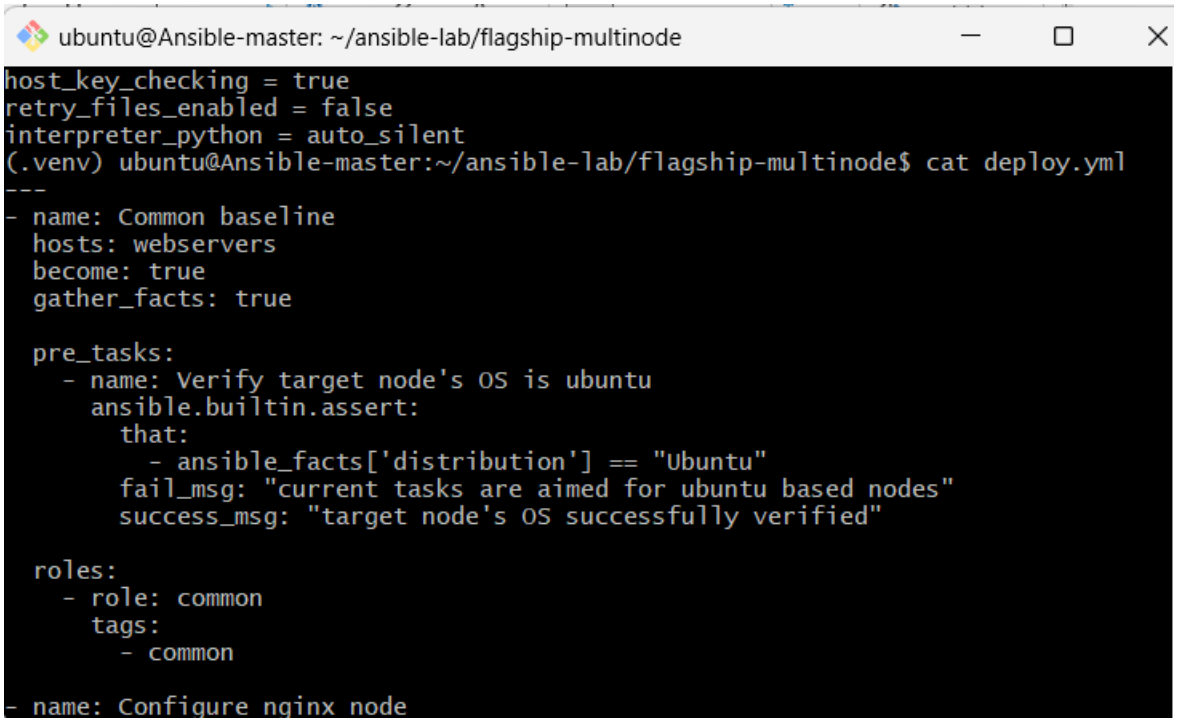



ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
./roles/common/defaults/main.yml
./roles/common/handlers
./roles/common/handlers/main.yml
./roles/apache
./roles/apache/tasks
./roles/apache/tasks/main.yml
./roles/apache/defaults
./roles/apache/defaults/main.yml
./roles/apache/templates
./roles/apache/templates/site.conf.j2
./roles/apache/templates/index.html.j2
./roles/apache/handlers
./roles/apache/handlers/check_reload.yml
./roles/apache/handlers/main.yml
./inventory
./inventory/hosts.ini
./ansible.cfg
./group_vars
./group_vars/web_nginx.yml
./group_vars/all.yml
./group_vars/web_apache.yml
./site.yml
./deploy.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$


ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat host_vars/web_01.yml
web_node_name: web_01
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat host_vars/web_02.yml
web_node_name: web_02
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat group_vars/all.yml
---
common_packages:
- curl
- chrony
common_timezone: Asia/Kolkata
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat group_vars/web_nginx.yml
---
web_server_type: nginx
web_server_package: nginx
web_server_service: nginx
web_document_root: /var/www/ansible-nginx
web_site_title: Ansible-Managed-Nginx


ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
web_listen_port: 80
web_environment: production
nginx_sites_available: /etc/nginx/sites-available/ansible-managed
nginx_sites_enabled: /etc/nginx/sites-enabled/ansible-managed
nginx_default_site: /etc/nginx/sites-enabled/default
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat group_vars/we
b_apache.yml
---
web_server_type: apache
web_server_package: apache2
web_server_service: apache2
web_document_root: /var/www/ansible-apache
web_site_title: Ansible-Managed-Apache
web_listen_port: 80
web_environment: production
apache_sites_available: /etc/apache2/sites-available/ansible-managed.conf
apache_sites_enabled: /etc/apache2/sites-enabled/ansible-managed.conf
apache_default_site: /etc/apache2/sites-enabled/000-default.conf
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat inventory/hos

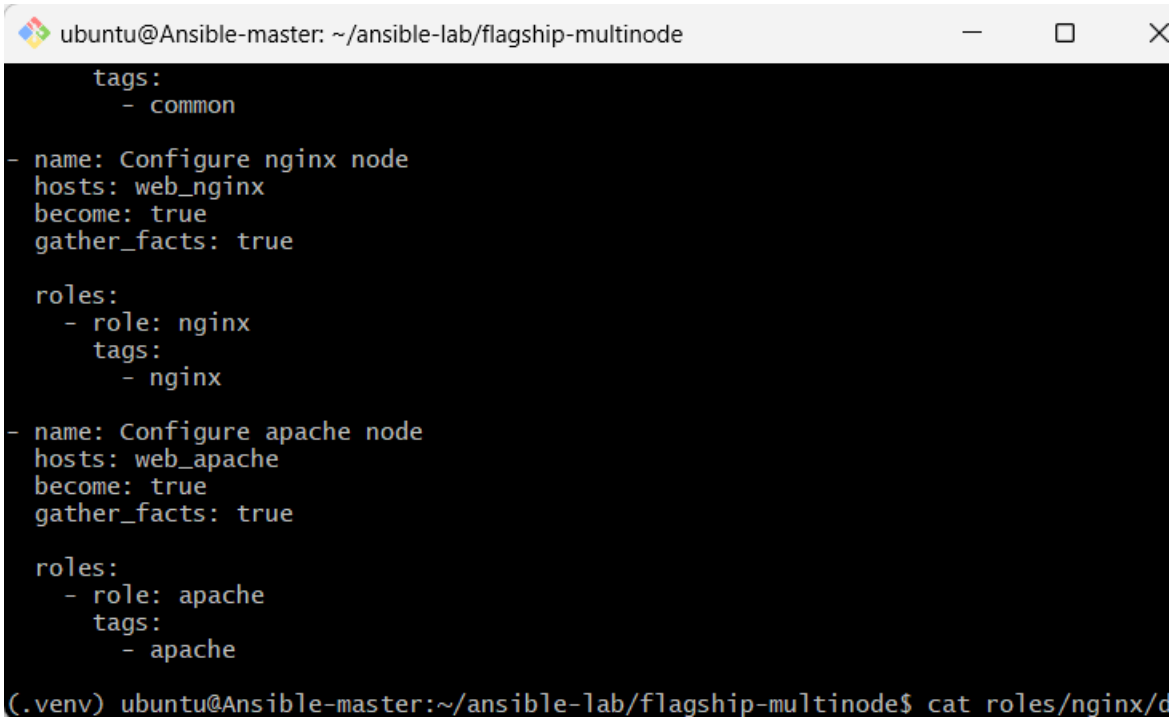

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat inventory/hos
ts.ini
[web_nginx]
web_01 ansible_host=172.31.14.107
[web_apache]
web_02 ansible_host=172.31.8.226
[webservers:children]
web_apache
web_nginx
[managed:children]
webservers
[managed:vars]
ansible_user=ubuntu
ansible_ssh_private_key_file=/home/ubuntu/.ssh/ansible_lab_key
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat ansible.cfg
[defaults]
inventory = ./inventory/hosts.ini
roles_path = ./roles
host_key_checking = true


ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
host_key_checking = true
retry_files_enabled = false
interpreter_python = auto_silent
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat deploy.yml
---
- name: Common baseline
  hosts: webservers
  become: true
  gather_facts: true

  pre_tasks:
    - name: Verify target node's OS is ubuntu
      ansible.builtin.assert:
        that:
          - ansible_facts['distribution'] == "Ubuntu"
        fail_msg: "current tasks are aimed for ubuntu based nodes"
        success_msg: "target node's OS successfully verified"

  roles:
    - role: common
      tags:
        - common

- name: Configure nginx node


ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
      tags:
        - common

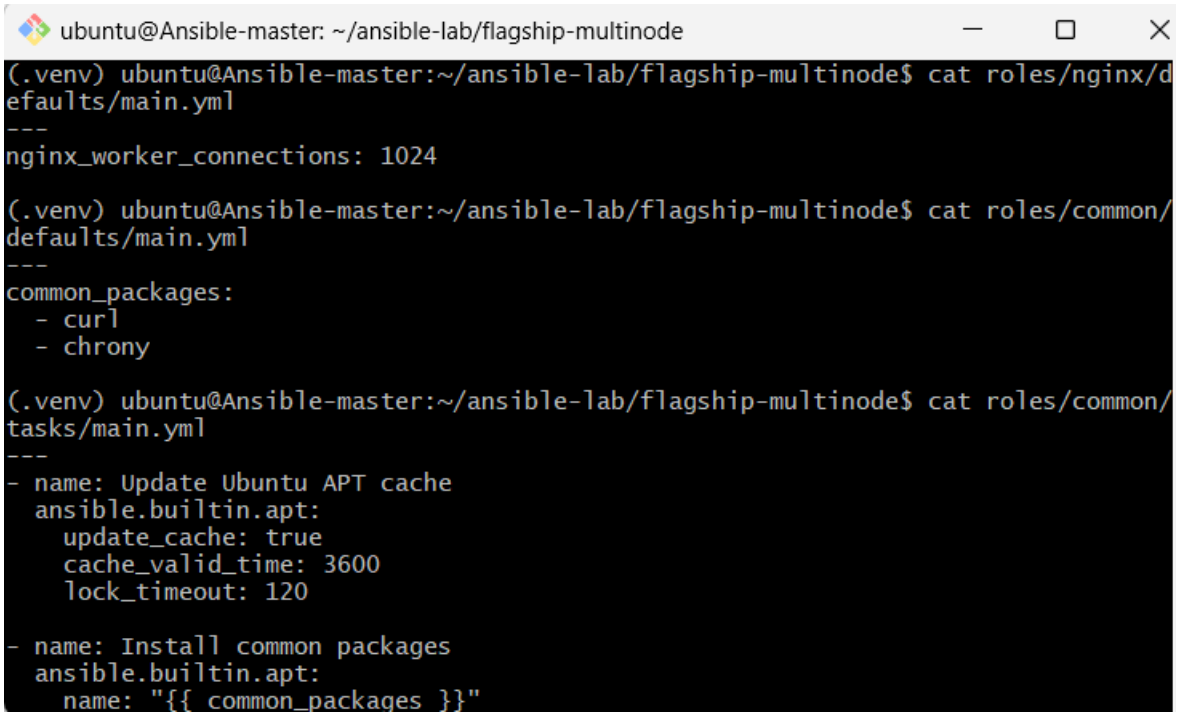
- name: Configure nginx node
  hosts: web_nginx
  become: true
  gather_facts: true

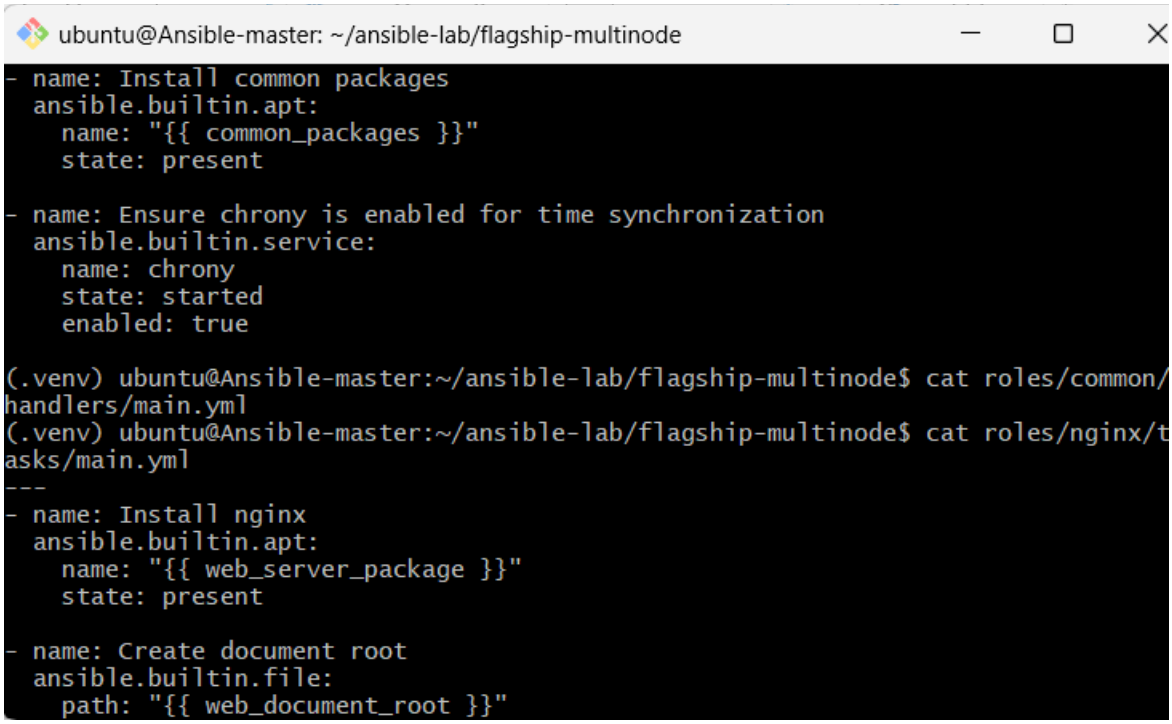
  roles:
    - role: nginx
      tags:
        - nginx

- name: Configure apache node
  hosts: web_apache
  become: true
  gather_facts: true

  roles:
    - role: apache
      tags:
        - apache

(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/nginx/d


ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/nginx/defaults/main.yml
---
nginx_worker_connections: 1024
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/common/defaults/main.yml
---
common_packages:
- curl
- chrony
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/common/tasks/main.yml
---
- name: Update Ubuntu APT cache
ansible.builtin.apt:
update_cache: true
cache_valid_time: 3600
lock_timeout: 120
- name: Install common packages
ansible.builtin.apt:
name: "{{ common_packages }}"


ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
- name: Install common packages
ansible.builtin.apt:
name: "{{ common_packages }}"
state: present
- name: Ensure chrony is enabled for time synchronization
ansible.builtin.service:
name: chrony
state: started
enabled: true
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/common/handlers/main.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/nginx/tasks/main.yml
---
- name: Install nginx
ansible.builtin.apt:
name: "{{ web_server_package }}"
state: present
- name: Create document root
ansible.builtin.file:
path: "{{ web_document_root }}"

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode

```
- name: Create document root
  ansible.builtin.file:
    path: "{{ web_document_root }}"
    state: directory
    owner: root
    group: root
    mode: "0755"

- name: Deploy index.html
  ansible.builtin.template:
    src: index.html.j2
    dest: "{{ web_document_root }}/index.html"
    owner: root
    group: root
    mode: "0644"
  notify: Check and reload nginx

- name: Deploy nginx site conf
  ansible.builtin.template:
    src: site.conf.j2
    dest: "{{ nginx_sites_available }}"
    owner: root
    group: root
    mode: "0644"
```

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode

```
- name: Deploy nginx site conf
  ansible.builtin.template:
    src: site.conf.j2
    dest: "{{ nginx_sites_available }}"
    owner: root
    group: root
    mode: "0644"
  notify: Check and reload nginx

- name: Disable nginx default site
  ansible.builtin.file:
    path: "{{ nginx_default_site }}"
    state: absent
  notify: Check and reload nginx

- name: Create sym-link
  ansible.builtin.file:
    src: "{{ nginx_sites_available }}"
    dest: "{{ nginx_sites_enabled }}"
    state: link
    owner: root
    group: root
  notify: Check and reload nginx
```

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode

```
- name: Ensure nginx is enabled and running
  ansible.builtin.service:
    name: nginx
    state: started
    enabled: true

- name: Apply pending configuration changes to the node
  ansible.builtin.meta: flush_handlers

- name: Ensure http endpoint is working
  ansible.builtin.uri:
    url: "http://127.0.0.1:{{ web_listen_port }}/health"
    status_code: 200
    return_content: true
  register: output
  retries: 5
  delay: 2
  until: output.status | default(0) == 200

- name: Display nginx deployment result
  ansible.builtin.debug:
    msg:
      - "nginx_hostname: {{ ansible_facts['hostname'] }}"
      - "status_code: {{ output.status }}"
```

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode

```
- name: Display nginx deployment result
  ansible.builtin.debug:
    msg:
      - "nginx_hostname: {{ ansible_facts['hostname'] }}"
      - "status_code: {{ output.status }}"
      - "content: {{ output.content | trim }}"
      - "server: nginx"

(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/nginx/t
emplates/index.html.j2
<h1>Multi-Node practical</h1>
<h2>Web site title: {{ web_site_title }}</h2>
<p>Web node name: {{ web_node_name }}</p>
<p>Server: {{ web_server_type }}</p>
<p>Environment: {{ web_environment }}</p>
<p>OS: {{ ansible_facts['distribution'] }},{{ ansible_facts['distribution_version
'] }}</p>
<h3>Manged by Ansible and Vishnu</h3>
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/nginx/t
emplates/site.conf.j2
server{
  listen {{ web_listen_port }} default_server;
  listen [::]:{{ web_listen_port }} default_server;
```

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode

```
server{
  listen {{ web_listen_port }} default_server;
  listen [::]:{{ web_listen_port }} default_server;

  server_name _;
  root {{ web_document_root }};
  index index.html;

  server_tokens off;

  location / {
    try_files $uri $uri/ =404;
  }

  location = /health {
    access_log off;
    default_type text/plain;
    return 200 "healthy\n";
  }

  location ~ /\. {
    deny all;
  }
}
```

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode

```
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/nginx/h
andlers/main.yml
---
- name: Check and reload nginx
  ansible.builtin.include_tasks: check_reload.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/nginx/h
andlers/check_reload.yml
---
- name: Validate nginx
  ansible.builtin.command:
    argv:
      - nginx
      - -t
  changed_when: false

- name: Reload nginx
  ansible.builtin.service:
    name: nginx
    state: reloaded

(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/apache/
defaults/main.yml
---
apache_server_tokens: Prod
```

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode

```
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/apache/
tasks/main.yml
---
- name: Install Ansible
  ansible.builtin.apt:
    name: "{{ web_server_package }}"
    state: present

- name: Create document root
  ansible.builtin.file:
    path: "{{ web_document_root }}"
    state: directory
    owner: root
    group: root
    mode: "0755"

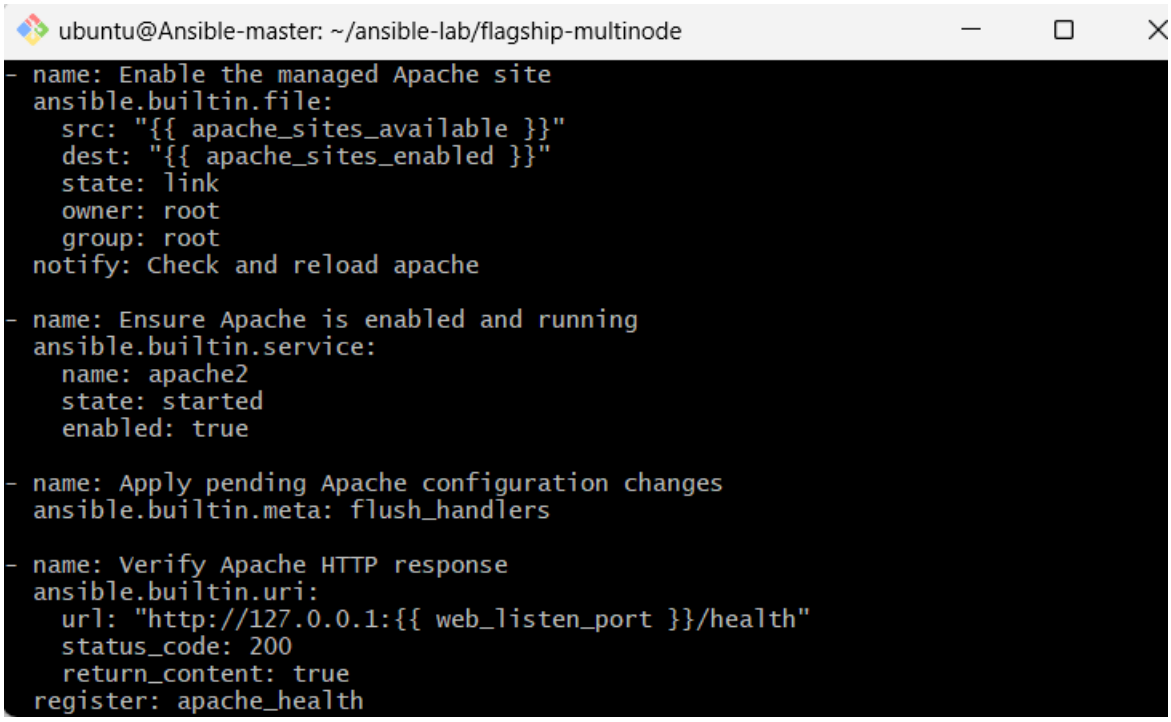
- name: Deploy Apache index page
  ansible.builtin.template:
    src: index.html.j2
    dest: "{{ web_document_root }}/index.html"
    owner: www-data
    group: www-data
    mode: "0644"
  notify: Check and reload apache
```

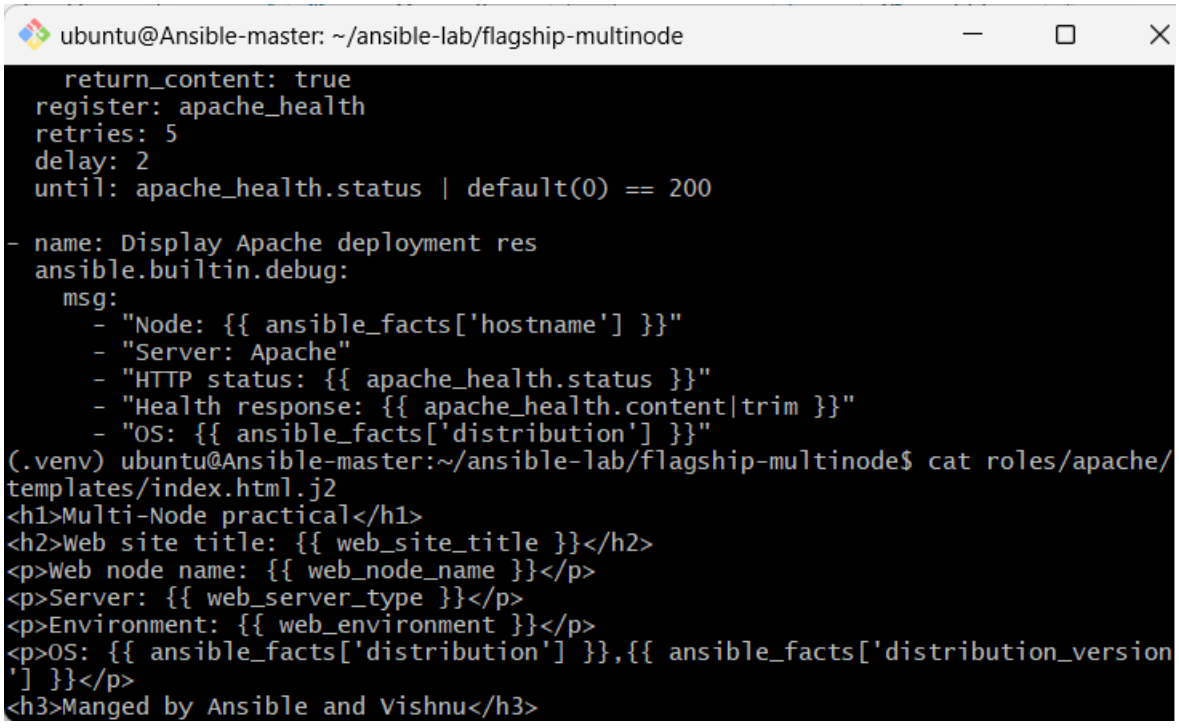
ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode

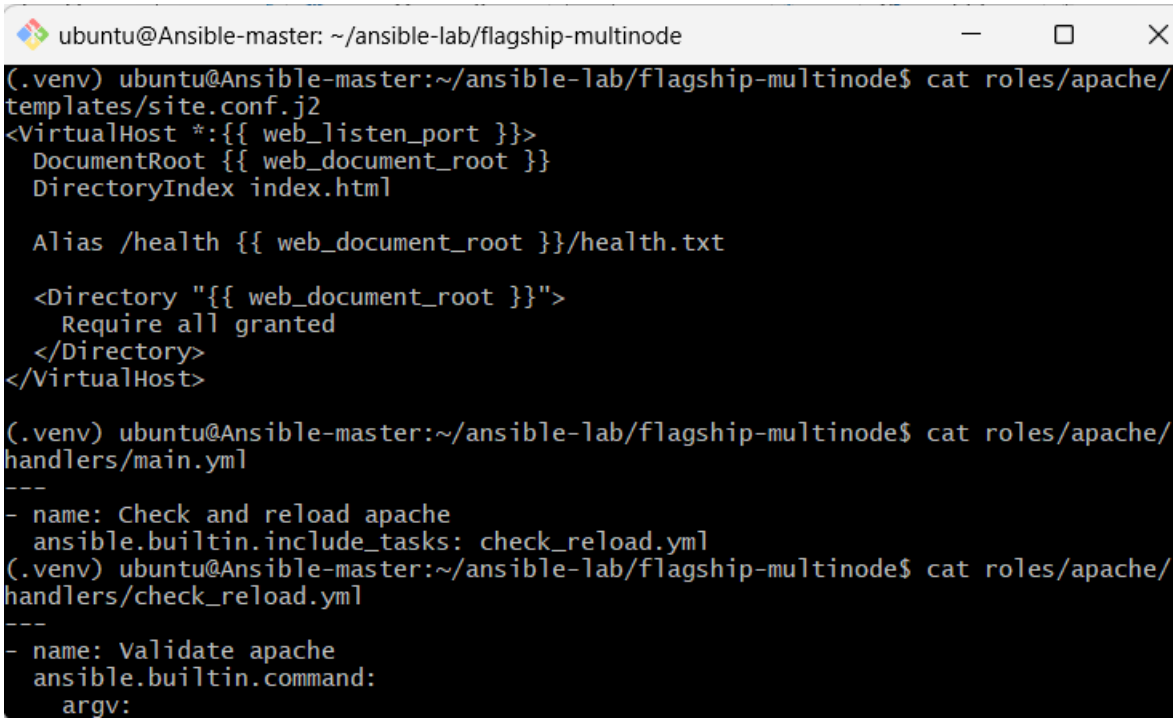
```
- name: Deploy Apache health endpoint file
  ansible.builtin.copy:
    content: "healthy-apache\n"
    dest: "{{ web_document_root }}/health.txt"
    owner: www-data
    group: www-data
    mode: "0644"
  notify: Check and reload apache

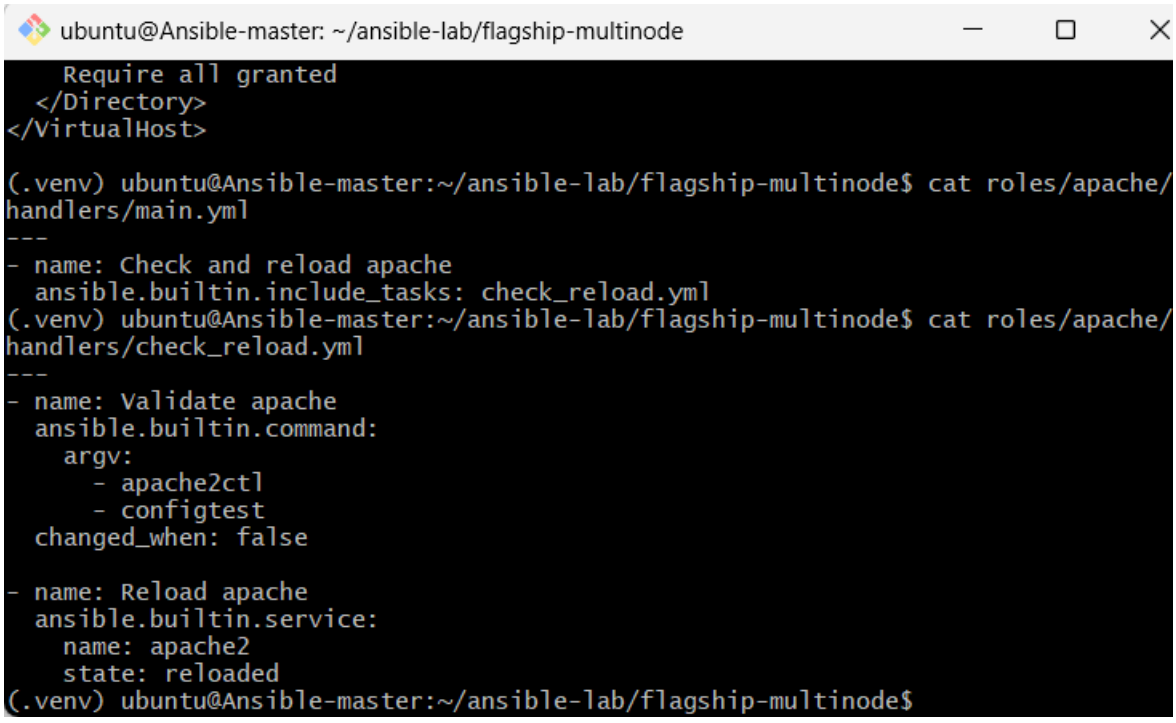
- name: Deploy Apache virtual host conf
  ansible.builtin.template:
    src: site.conf.j2
    dest: "{{ apache_sites_available }}"
    owner: root
    group: root
    mode: "0644"
  notify: Check and reload apache

- name: Disable the default apache site
  ansible.builtin.file:
    path: "{{ apache_default_site }}"
    state: absent
  notify: Check and reload apache
```

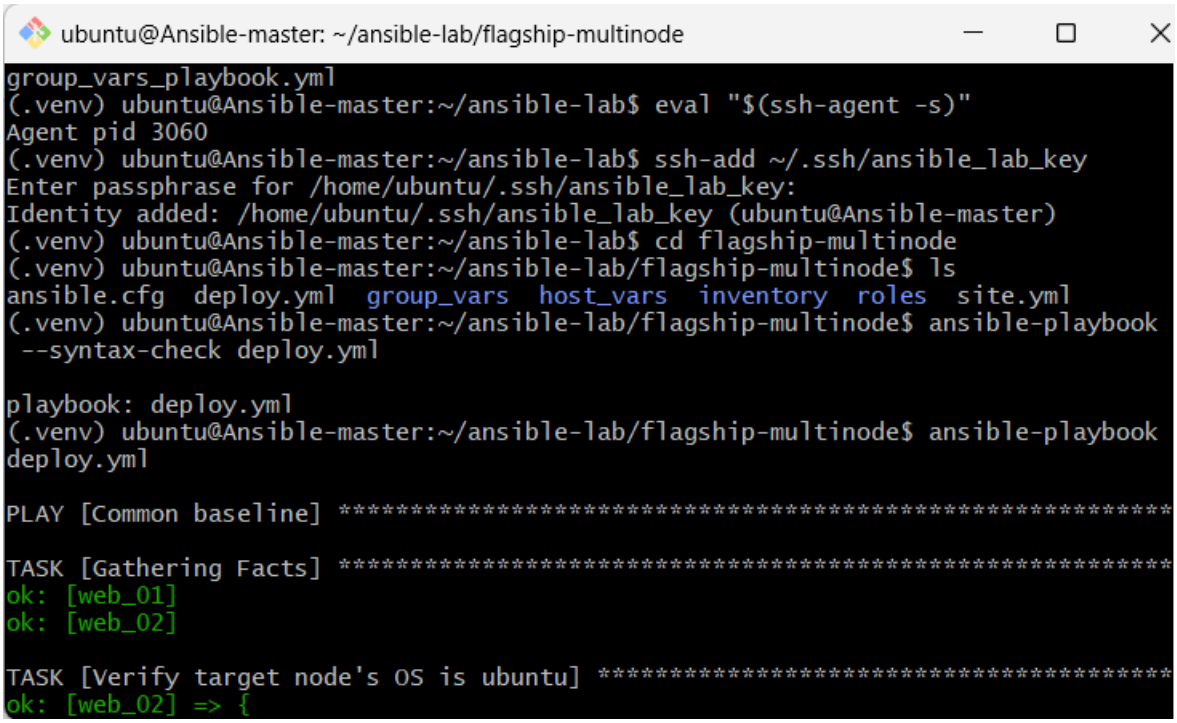

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
- name: Enable the managed Apache site
ansible.builtin.file:
src: "{{ apache_sites_available }}"
dest: "{{ apache_sites_enabled }}"
state: link
owner: root
group: root
notify: Check and reload apache
- name: Ensure Apache is enabled and running
ansible.builtin.service:
name: apache2
state: started
enabled: true
- name: Apply pending Apache configuration changes
ansible.builtin.meta: flush_handlers
- name: Verify Apache HTTP response
ansible.builtin.uri:
url: "http://127.0.0.1:{{ web_listen_port }}/health"
status_code: 200
return_content: true
register: apache_health

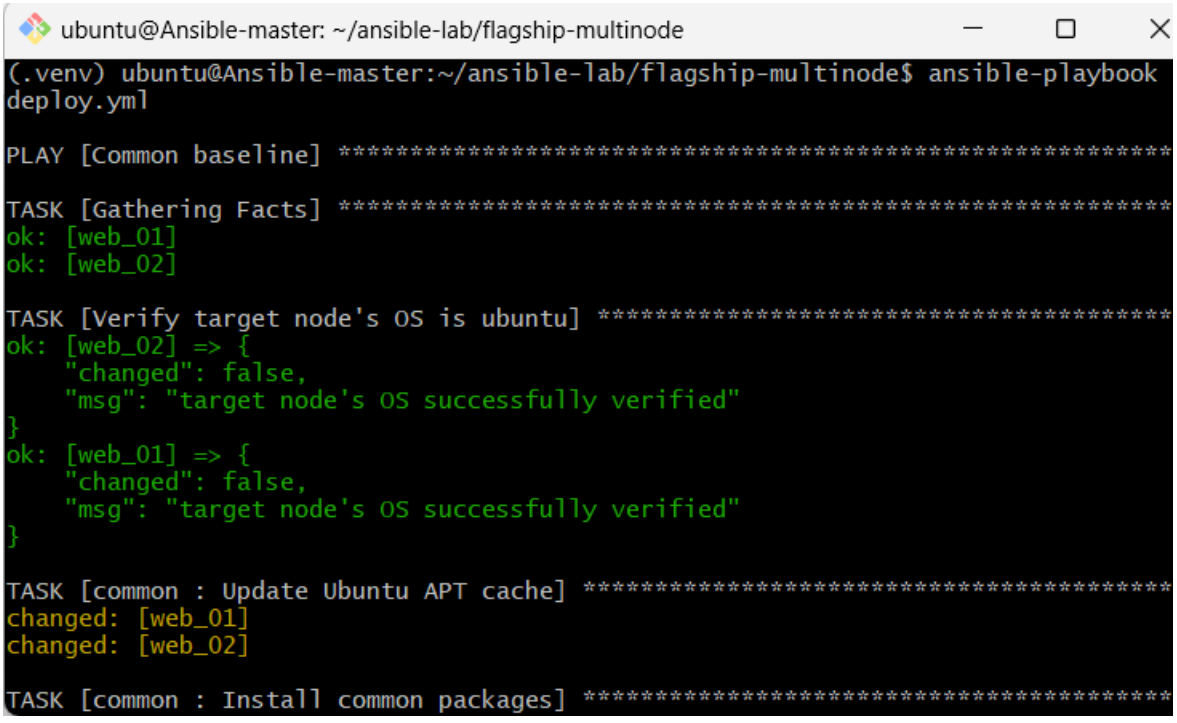

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
return_content: true
register: apache_health
retries: 5
delay: 2
until: apache_health.status | default(0) == 200
- name: Display Apache deployment res
ansible.builtin.debug:
msg:
- "Node: {{ ansible_facts['hostname'] }}"
- "Server: Apache"
- "HTTP status: {{ apache_health.status }}"
- "Health response: {{ apache_health.content|trim }}"
- "OS: {{ ansible_facts['distribution'] }}"
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/apache/
templates/index.html.j2
<h1>Multi-Node practical</h1>
<h2>Web site title: {{ web_site_title }}</h2>
Web node name: {{ web_node_name }}
Server: {{ web_server_type }}
Environment: {{ web_environment }}
OS: {{ ansible_facts['distribution'] }},{{ ansible_facts['distribution_version
'] }}
<h3>Manged by Ansible and Vishnu</h3>

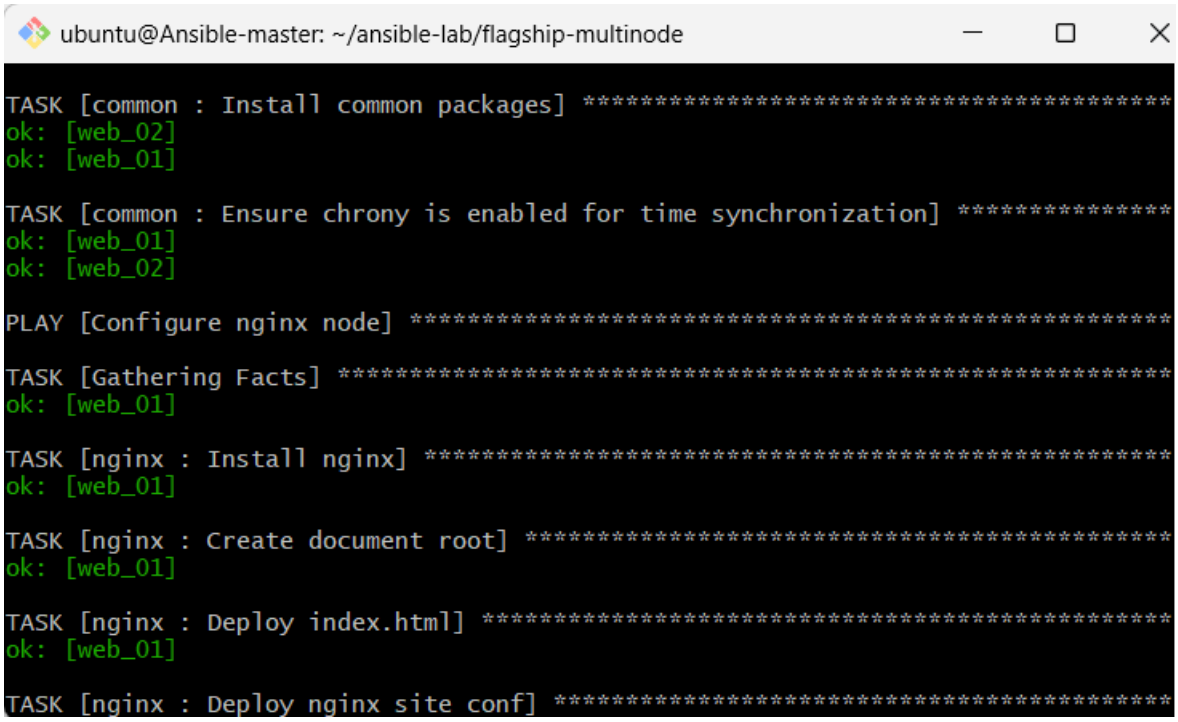

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/apache/templates/site.conf.j2
<VirtualHost *:{{ web_listen_port }}>
DocumentRoot {{ web_document_root }}
DirectoryIndex index.html
Alias /health {{ web_document_root }}/health.txt
<Directory "{{ web_document_root }}">
Require all granted
</Directory>
</VirtualHost>
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/apache/handlers/main.yml
---
- name: Check and reload apache
ansible.builtin.include_tasks: check_reload.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/apache/handlers/check_reload.yml
---
- name: Validate apache
ansible.builtin.command:
argv:

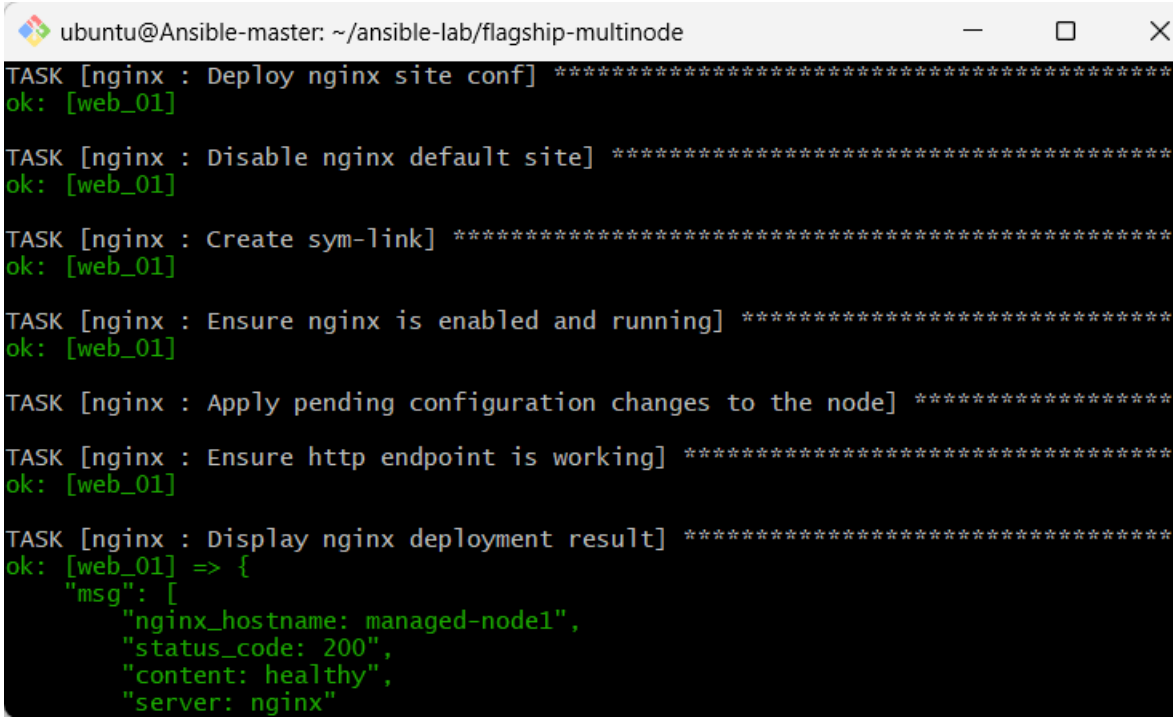

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
Require all granted
</Directory>
</VirtualHost>
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/apache/handlers/main.yml
---
- name: Check and reload apache
ansible.builtin.include_tasks: check_reload.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ cat roles/apache/handlers/check_reload.yml
---
- name: Validate apache
ansible.builtin.command:
argv:
- apache2ctl
- configtest
changed_when: false
- name: Reload apache
ansible.builtin.service:
name: apache2
state: reloaded
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$

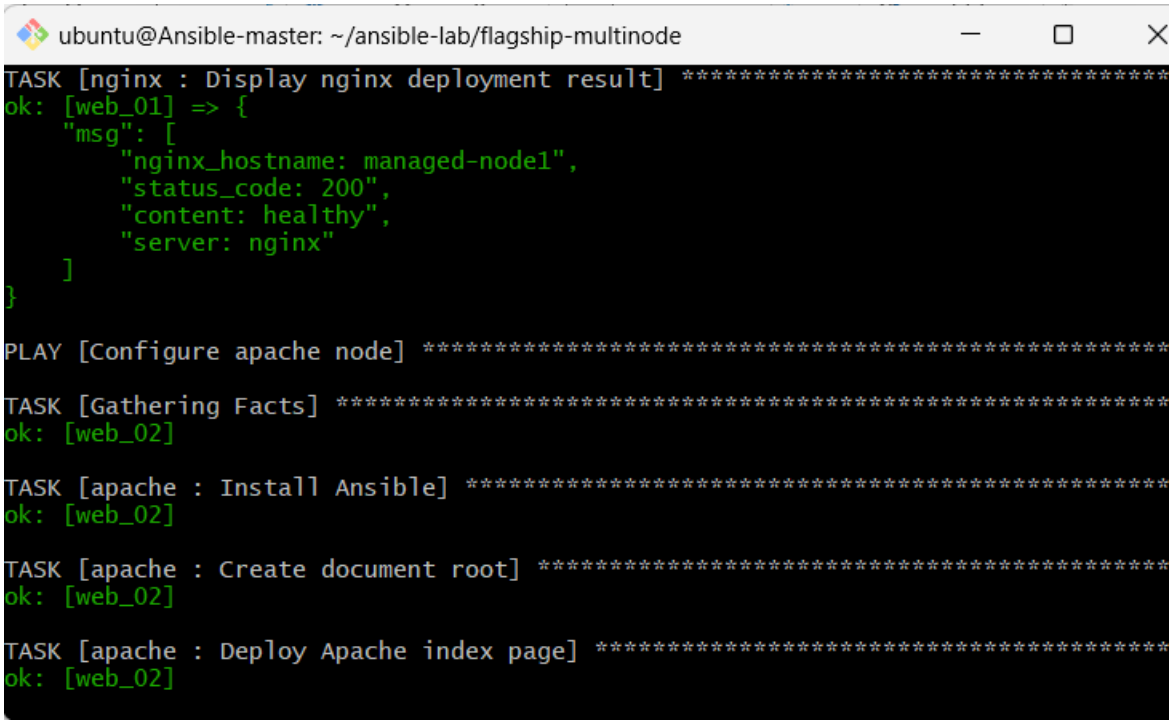
That was about the complete code part now comes the execution of this beautiful and extensive practical:

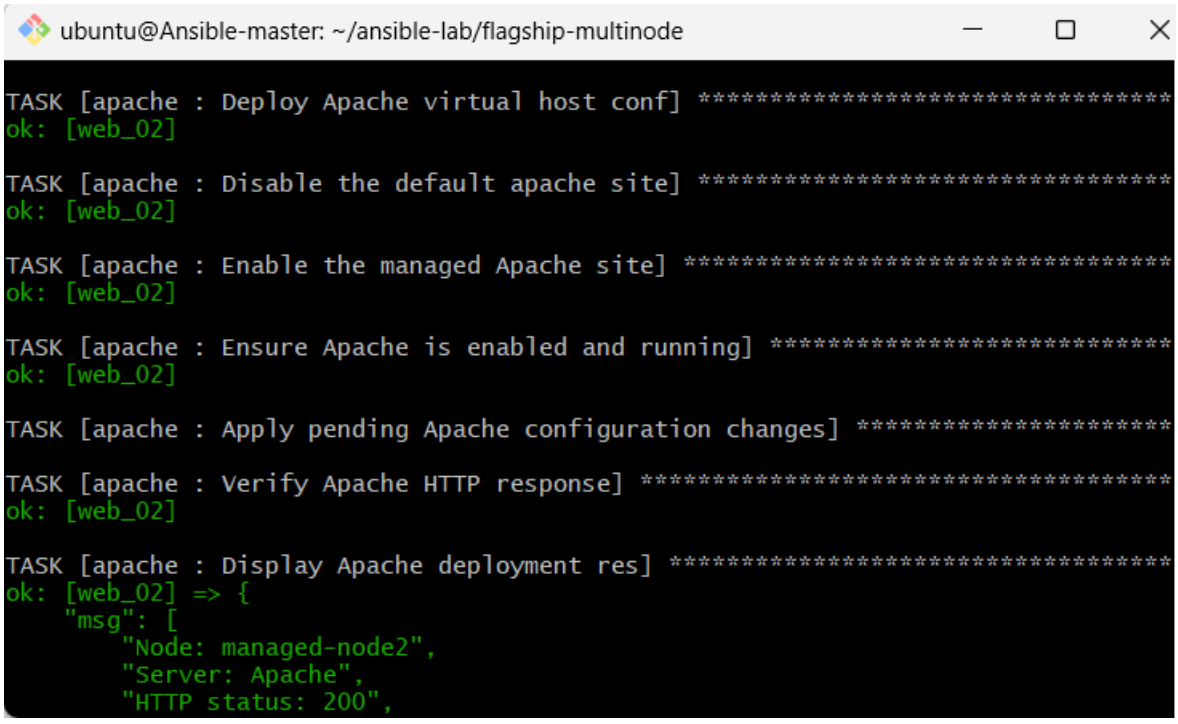


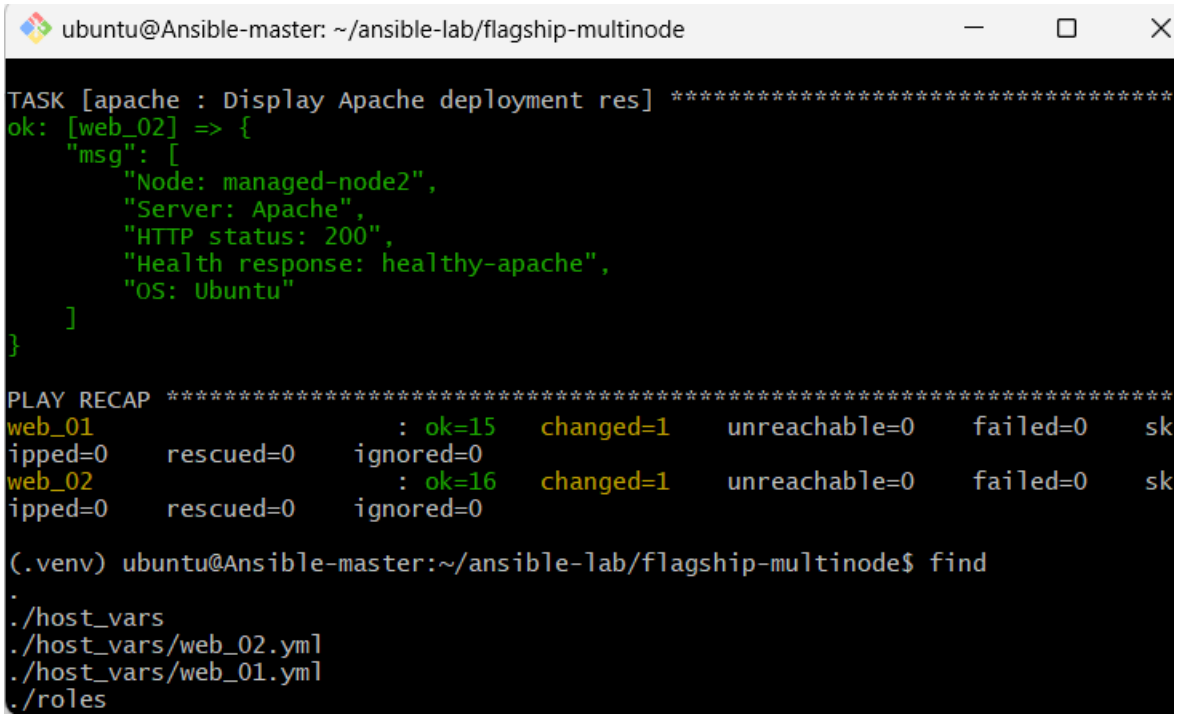


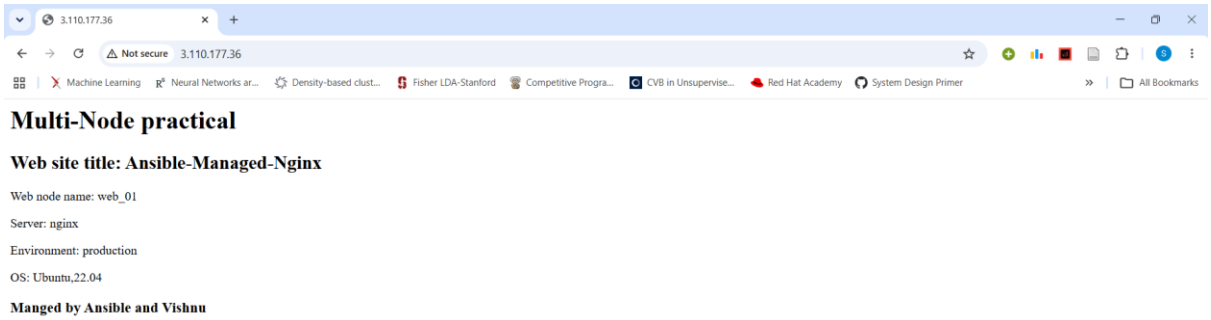

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
TASK [common : Install common packages] **
ok: [web_02]
ok: [web_01]
TASK [common : Ensure chrony is enabled for time synchronization] ***************
ok: [web_01]
ok: [web_02]
PLAY [Configure nginx node] **
TASK [Gathering Facts] ***
ok: [web_01]
TASK [nginx : Install nginx] ***
ok: [web_01]
TASK [nginx : Create document root] **
ok: [web_01]
TASK [nginx : Deploy index.html] ***
ok: [web_01]
TASK [nginx : Deploy nginx site conf] **


ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
TASK [nginx : Deploy nginx site conf] **
ok: [web_01]
TASK [nginx : Disable nginx default site] **************************************
ok: [web_01]
TASK [nginx : Create sym-link] ***
ok: [web_01]
TASK [nginx : Ensure nginx is enabled and running] *****************************
ok: [web_01]
TASK [nginx : Apply pending configuration changes to the node] *****************
TASK [nginx : Ensure http endpoint is working] *********************************
ok: [web_01]
TASK [nginx : Display nginx deployment result] *********************************
ok: [web_01] => {
"msg": [
"nginx_hostname: managed-node1",
"status_code: 200",
"content: healthy",
"server: nginx"

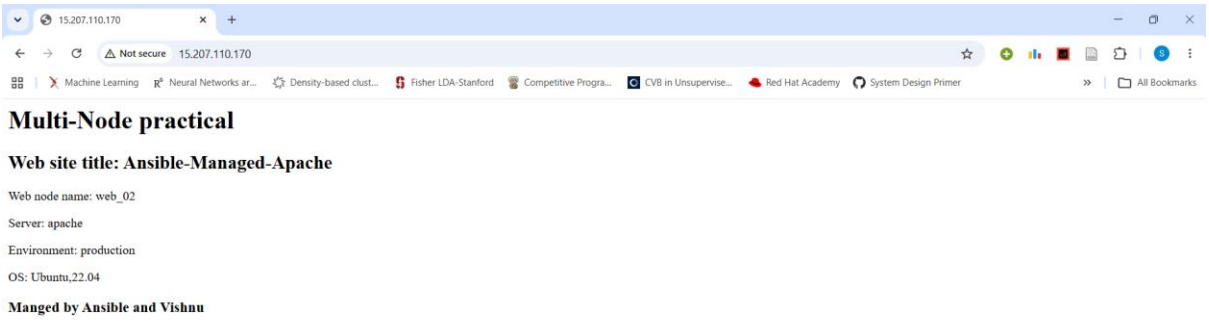

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
TASK [nginx : Display nginx deployment result] ********************************
ok: [web_01] => {
"msg": [
"nginx_hostname: managed-node1",
"status_code: 200",
"content: healthy",
"server: nginx"
]
}
PLAY [Configure apache node] ***
TASK [Gathering Facts] ***
ok: [web_02]
TASK [apache : Install Ansible] **
ok: [web_02]
TASK [apache : Create document root] ***
ok: [web_02]
TASK [apache : Deploy Apache index page] ***************************************
ok: [web_02]


ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
TASK [apache : Deploy Apache virtual host conf] ********************************
ok: [web_02]
TASK [apache : Disable the default apache site] ********************************
ok: [web_02]
TASK [apache : Enable the managed Apache site] *********************************
ok: [web_02]
TASK [apache : Ensure Apache is enabled and running] ***************************
ok: [web_02]
TASK [apache : Apply pending Apache configuration changes] *********************
TASK [apache : Verify Apache HTTP response] ************************************
ok: [web_02]
TASK [apache : Display Apache deployment res] **********************************
ok: [web_02] => {
"msg": [
"Node: managed-node2",
"Server: Apache",
"HTTP status: 200",



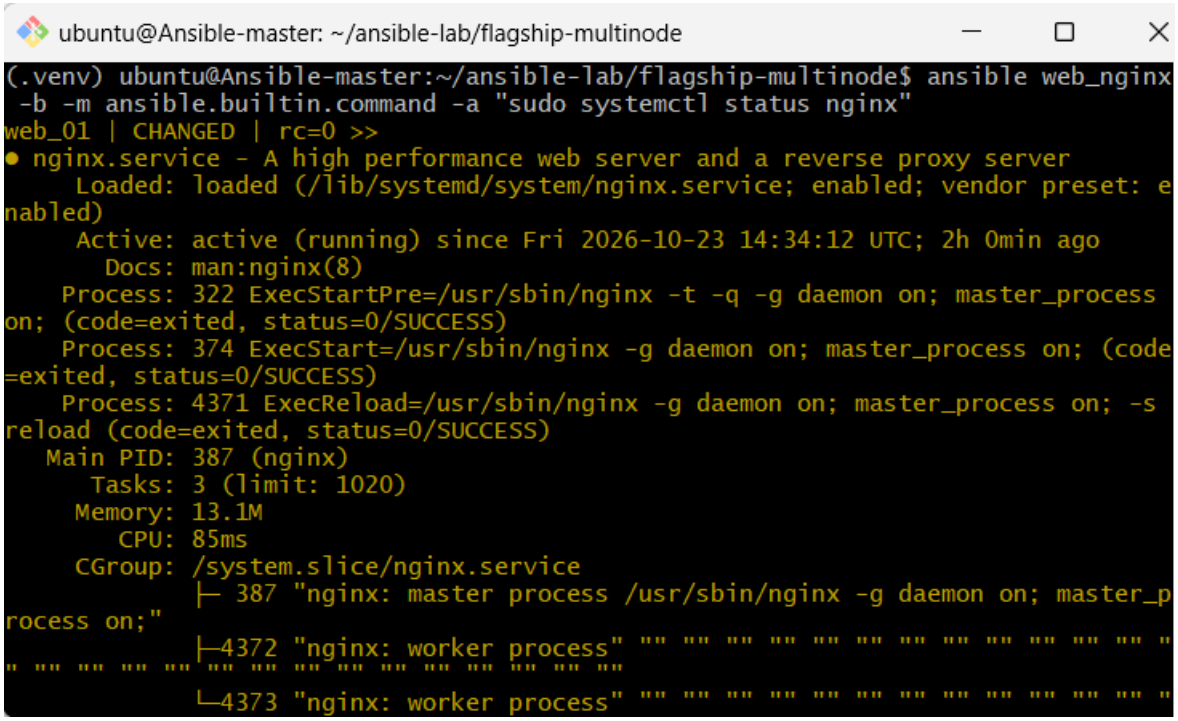


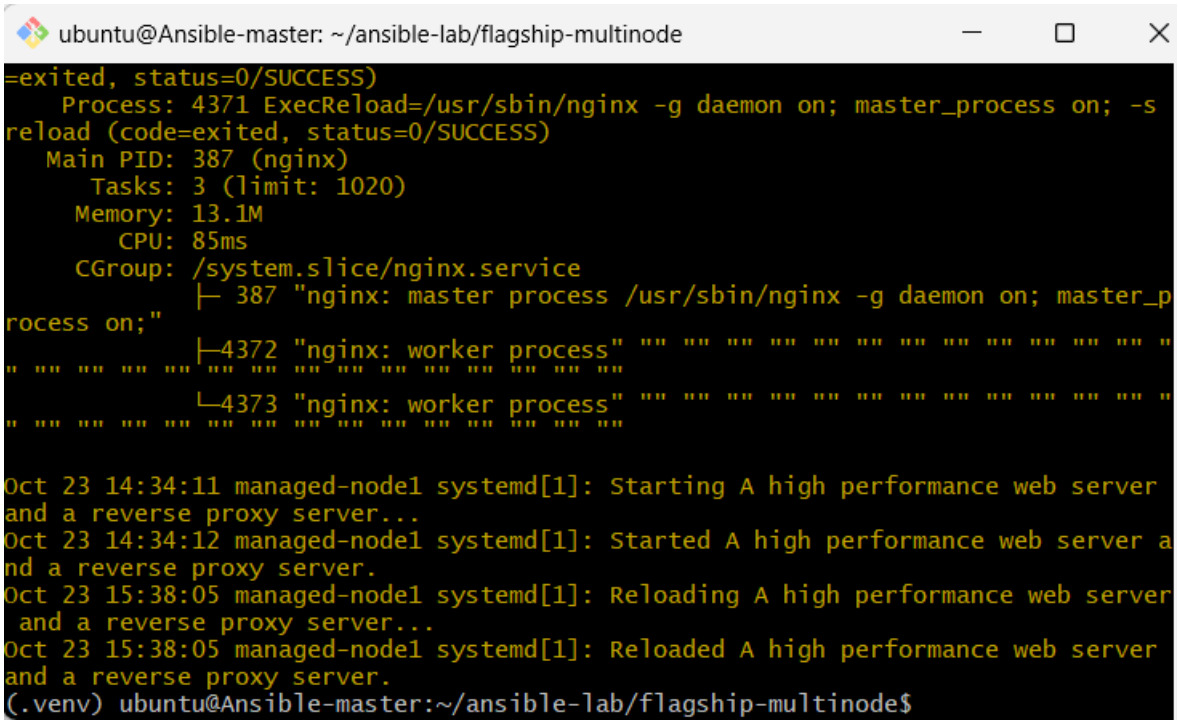
Nginx serves from the Manged-node1

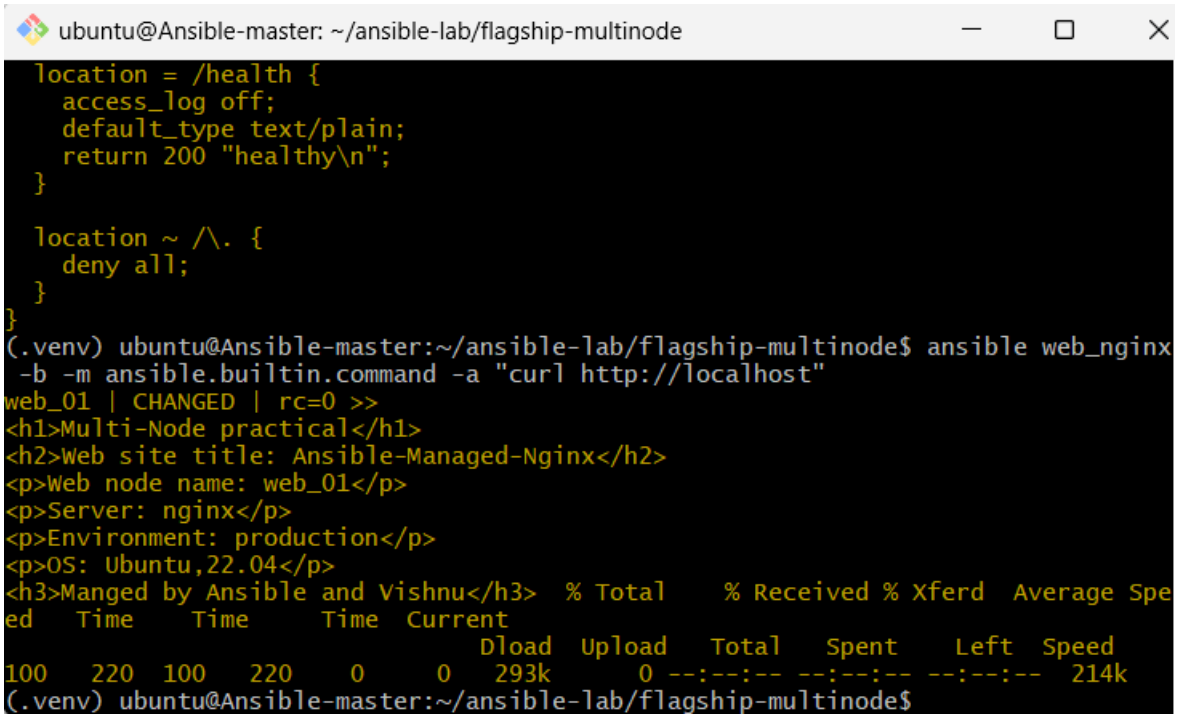


And there we go, Apache serves from the Managed-node2

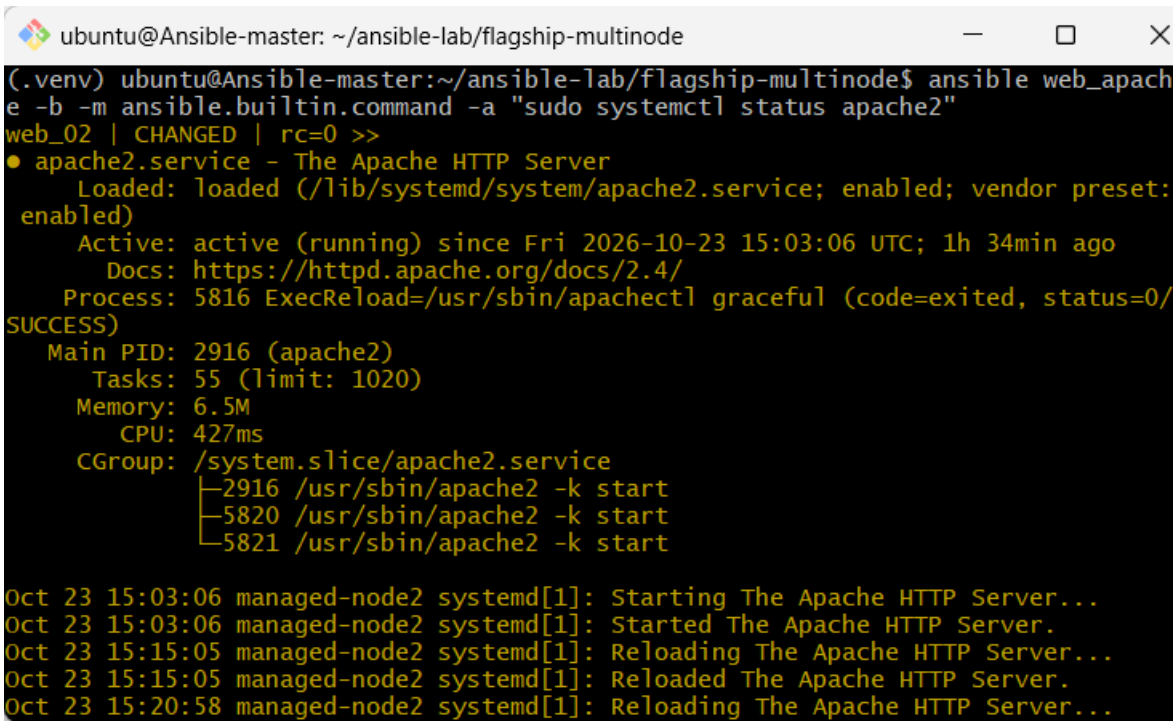
Confirmation of nginx status in managed-node1

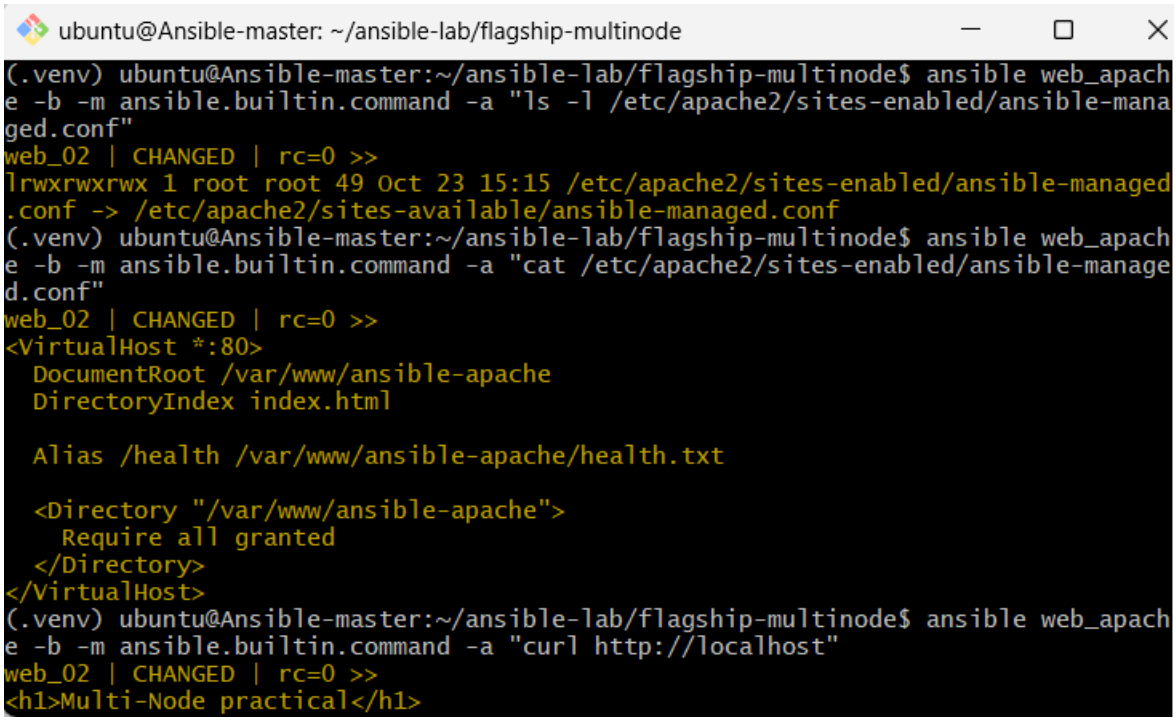


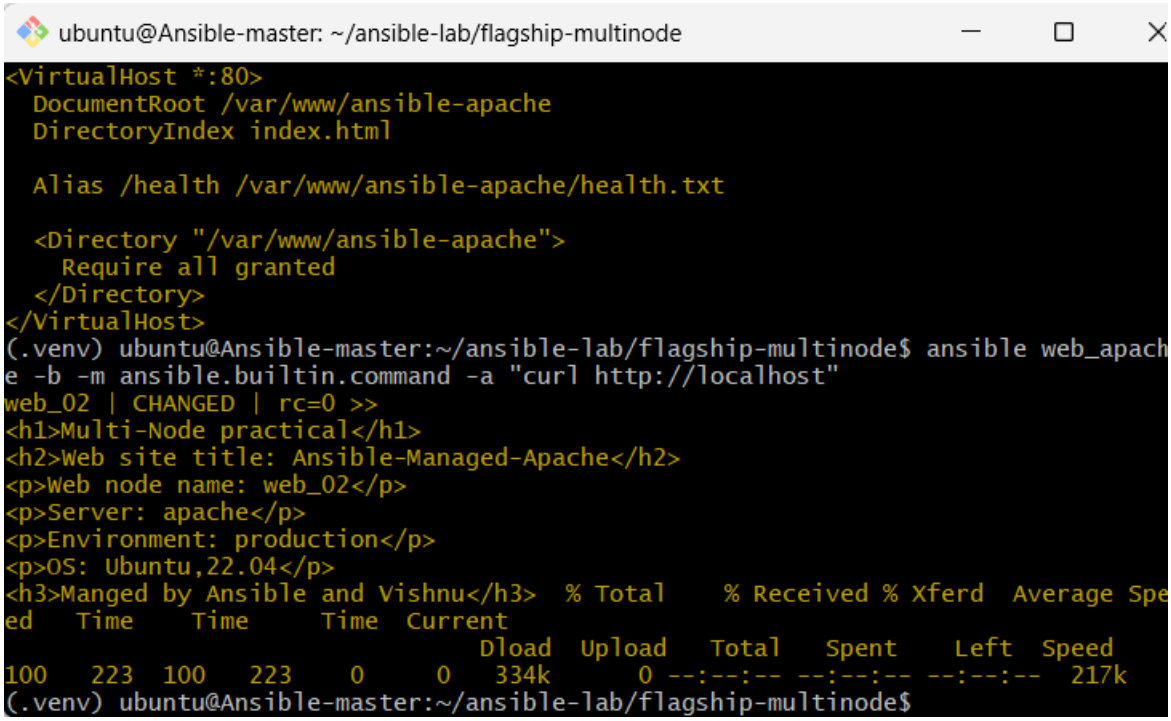

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
=exited, status=0/SUCCESS)
Process: 4371 ExecReload=/usr/sbin/nginx -g daemon on; master_process on; -s reload (code=exited, status=0/SUCCESS)
Main PID: 387 (nginx)
Tasks: 3 (limit: 1020)
Memory: 13.1M
CPU: 85ms
CGroup: /system.slice/nginx.service
├─ 387 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
├─4372 "nginx: worker process" "" "" "



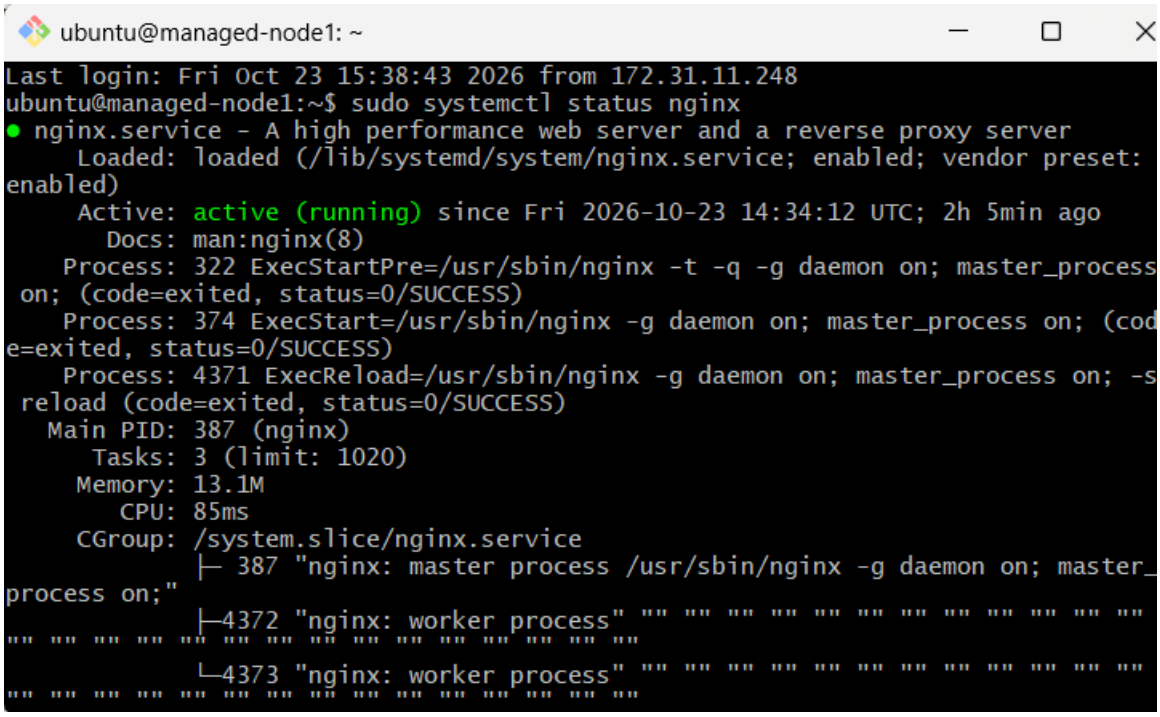
Confirmation of Apache status in managed-node2:

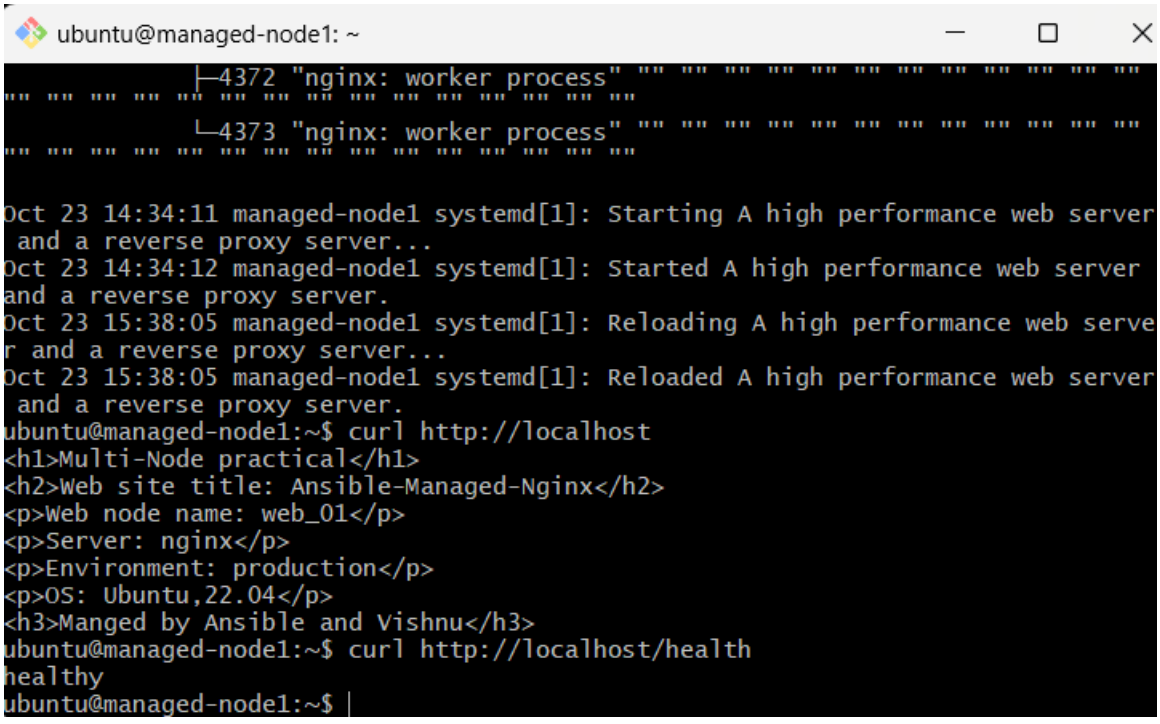


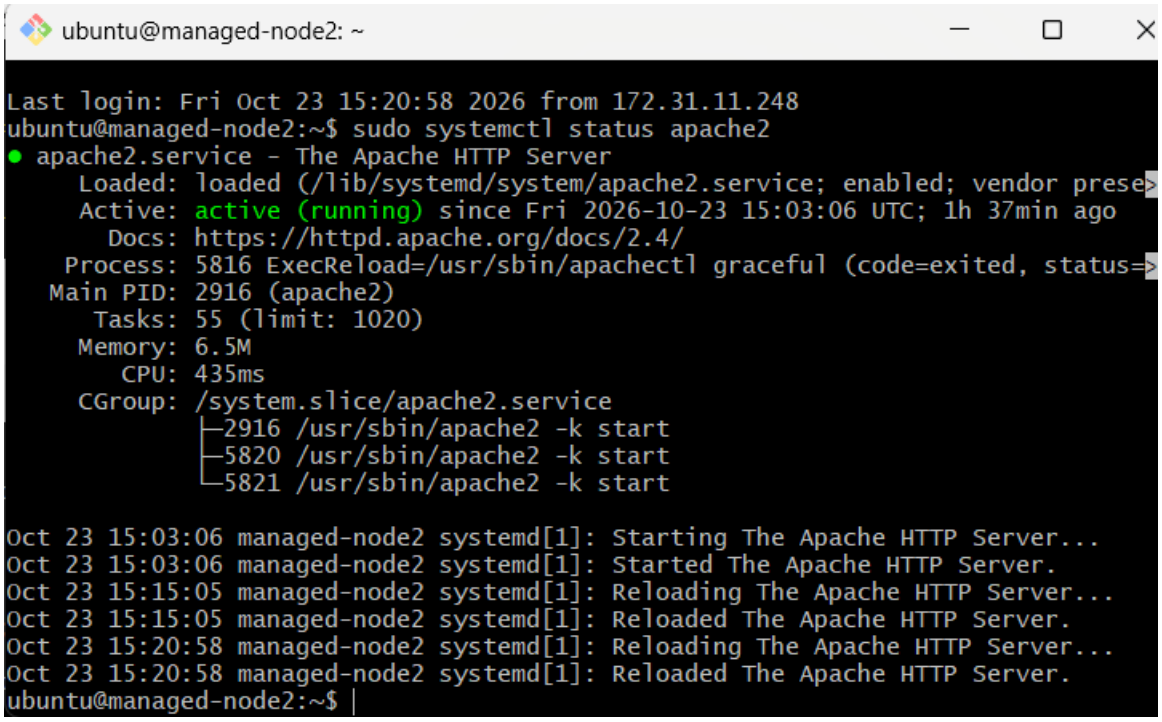

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ ansible web_apach
e -b -m ansible.builtin.command -a "ls -l /etc/apache2/sites-enabled/ansible-mana
ged.conf"
web_02 | CHANGED | rc=0 >>
lrwxrwxrwx 1 root root 49 Oct 23 15:15 /etc/apache2/sites-enabled/ansible-managed
.conf -> /etc/apache2/sites-available/ansible-managed.conf
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ ansible web_apach
e -b -m ansible.builtin.command -a "cat /etc/apache2/sites-enabled/ansible-manage
d.conf"
web_02 | CHANGED | rc=0 >>
<VirtualHost *:80>
DocumentRoot /var/www/ansible-apache
DirectoryIndex index.html
Alias /health /var/www/ansible-apache/health.txt
<Directory "/var/www/ansible-apache">
Require all granted
</Directory>
</VirtualHost>
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ ansible web_apach
e -b -m ansible.builtin.command -a "curl http://localhost"
web_02 | CHANGED | rc=0 >>
<h1>Multi-Node practical</h1>

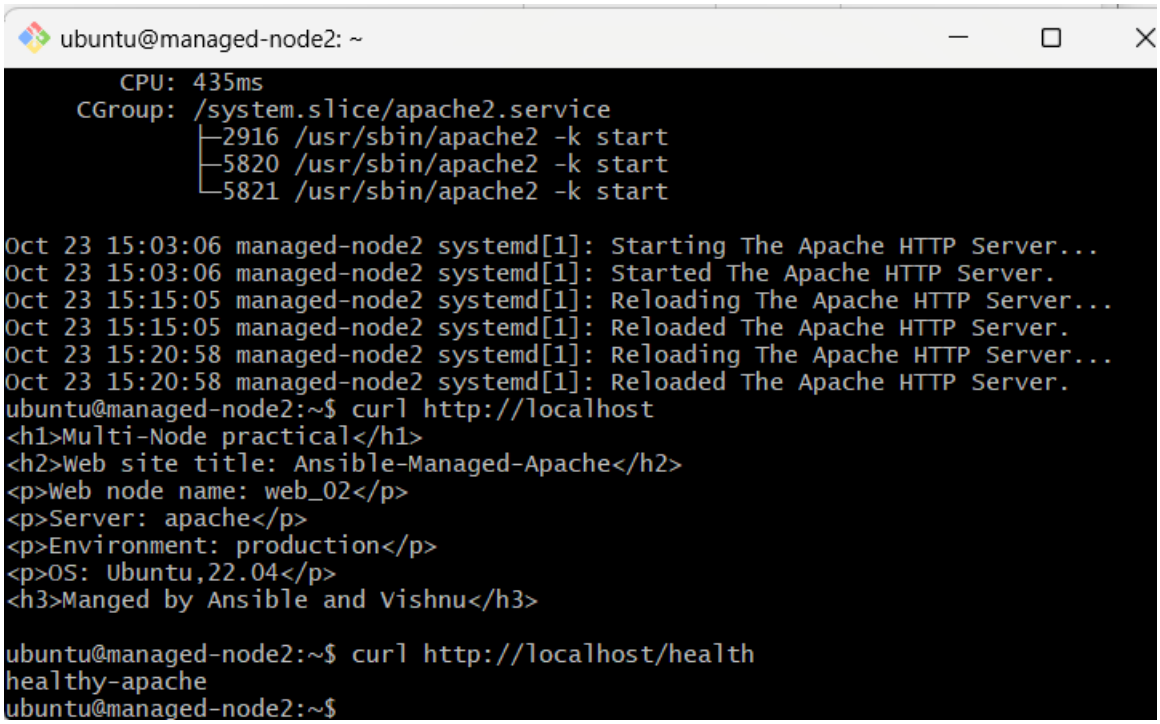

ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
<VirtualHost *:80>
DocumentRoot /var/www/ansible-apache
DirectoryIndex index.html
Alias /health /var/www/ansible-apache/health.txt
<Directory "/var/www/ansible-apache">
Require all granted
</Directory>
</VirtualHost>
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$ ansible web_apach
e -b -m ansible.builtin.command -a "curl http://localhost"
web_02 | CHANGED | rc=0 >>
<h1>Multi-Node practical</h1>
<h2>Web site title: Ansible-Managed-Apache</h2>
Web node name: web_02
Server: apache
Environment: production
OS: Ubuntu,22.04
<h3>Manged by Ansible and Vishnu</h3> % Total % Received % Xferd Average Spe
ed Time Time Time Current
Dload Upload Total Spent Left Speed
100 223 100 223 0 0 334k 0 --:--:-- --:--:-- --:--:-- 217k
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$

Now Ansible rocked, so let's do a bit of individual verification too:



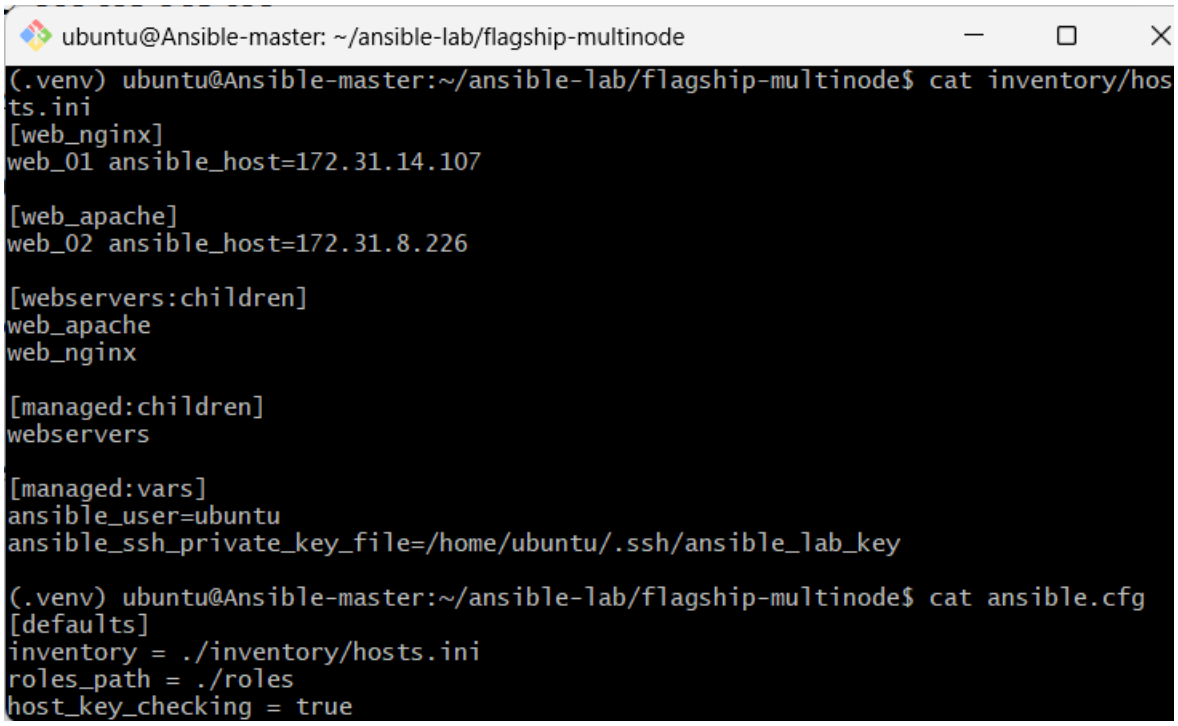


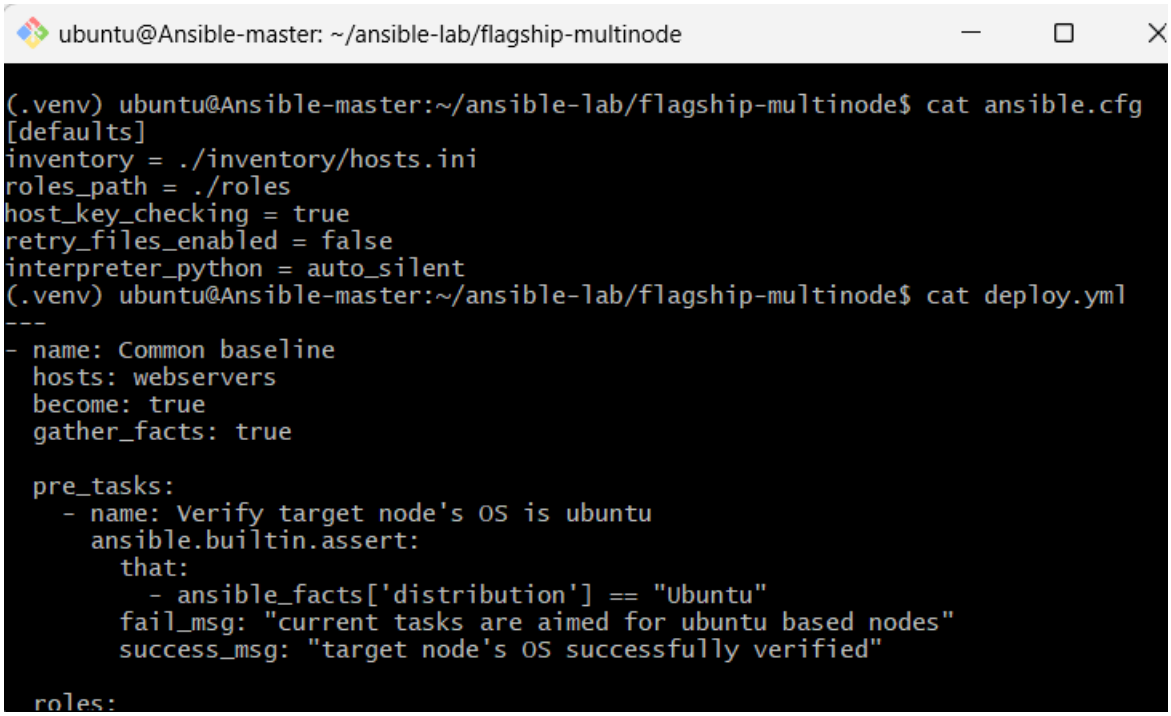

ubuntu@managed-node2: ~
Last login: Fri Oct 23 15:20:58 2026 from 172.31.11.248
ubuntu@managed-node2:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor prese
Active: active (running) since Fri 2026-10-23 15:03:06 UTC; 1h 37min ago
Docs: https://httpd.apache.org/docs/2.4/
Process: 5816 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=
Main PID: 2916 (apache2)
Tasks: 55 (limit: 1020)
Memory: 6.5M
CPU: 435ms
CGroup: /system.slice/apache2.service
├─2916 /usr/sbin/apache2 -k start
├─5820 /usr/sbin/apache2 -k start
└─5821 /usr/sbin/apache2 -k start
Oct 23 15:03:06 managed-node2 systemd[1]: Starting The Apache HTTP Server...
Oct 23 15:03:06 managed-node2 systemd[1]: Started The Apache HTTP Server.
Oct 23 15:15:05 managed-node2 systemd[1]: Reloading The Apache HTTP Server...
Oct 23 15:15:05 managed-node2 systemd[1]: Reloaded The Apache HTTP Server.
Oct 23 15:20:58 managed-node2 systemd[1]: Reloading The Apache HTTP Server...
Oct 23 15:20:58 managed-node2 systemd[1]: Reloaded The Apache HTTP Server.
ubuntu@managed-node2:~$

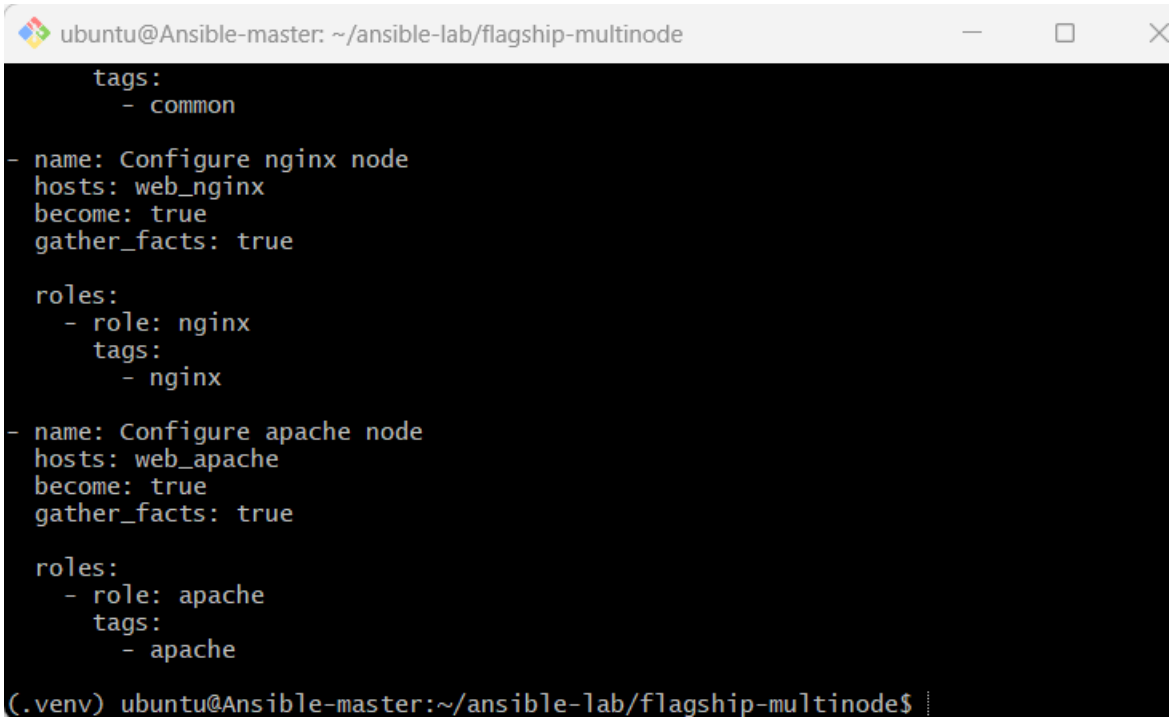

ubuntu@managed-node2: ~
CPU: 435ms
CGroup: /system.slice/apache2.service
├─2916 /usr/sbin/apache2 -k start
├─5820 /usr/sbin/apache2 -k start
└─5821 /usr/sbin/apache2 -k start
Oct 23 15:03:06 managed-node2 systemd[1]: Starting The Apache HTTP Server...
Oct 23 15:03:06 managed-node2 systemd[1]: Started The Apache HTTP Server.
Oct 23 15:15:05 managed-node2 systemd[1]: Reloading The Apache HTTP Server...
Oct 23 15:15:05 managed-node2 systemd[1]: Reloaded The Apache HTTP Server.
Oct 23 15:20:58 managed-node2 systemd[1]: Reloading The Apache HTTP Server...
Oct 23 15:20:58 managed-node2 systemd[1]: Reloaded The Apache HTTP Server.
ubuntu@managed-node2:~$ curl http://localhost
<h1>Multi-Node practical</h1>
<h2>Web site title: Ansible-Managed-Apache</h2>
Web node name: web_02
Server: apache
Environment: production
OS: Ubuntu,22.04
<h3>Manged by Ansible and Vishnu</h3>
ubuntu@managed-node2:~$ curl http://localhost/health
healthy-apache
ubuntu@managed-node2:~$

What better!!

Main files (a recap for fast revision):






ubuntu@Ansible-master: ~/ansible-lab/flagship-multinode
tags:
- common
- name: Configure nginx node
hosts: web_nginx
become: true
gather_facts: true
roles:
- role: nginx
tags:
- nginx
- name: Configure apache node
hosts: web_apache
become: true
gather_facts: true
roles:
- role: apache
tags:
- apache
(.venv) ubuntu@Ansible-master:~/ansible-lab/flagship-multinode$